

Dairy Farm Water, Energy and Feed Reference

100-Cow Lactating Herd | Texas Climate | System Dynamics Model

1. Data Collection: Water and Energy Use

Water Use

Parameter	Value	Unit	Source
Drinking water per cow	150	L/cow/day	Penn State Extension
Total - 100 cows	15,000	L/day	150 x 100
Total farm (all uses)	220,000-250,000	L/month	USDA NRCS
Precipitation (College Station)	75	mm/month	NOAA 30-yr normals
Groundwater recharge	15	mm/month	~20% of precipitation
Wastewater recycling efficiency	75%	N/A	EPA lagoon standard

Energy Use

Parameter	Value	Unit	Source
Energy use per cow	60-70	kWh/cow/month	USDA ERS 2023
Total - 100 cows	6,500	kWh/month	65 kWh x 100
Solar PV (100 kW array)	~850	kWh/month	NREL PVWatts TX
Biogas from manure	~5	kWh/m3 CH4	EPA AgSTAR
Renewable offset	35-40%	of total demand	Solar + biogas combined

Sources: Penn State Extension; USDA NRCS; NOAA 1991-2020; USDA ERS 2023; NREL PVWatts v8; EPA AgSTAR.

2. Feed for Lactating Dairy Cows

Lactating cows require a high-energy Total Mixed Ration (TMR). A typical ration for a Texas herd includes corn silage as the main energy base (40-50% of dry matter), alfalfa hay for protein and fiber (15-25%), and bermudagrass hay as a local forage option (5-10%). Ground corn adds starch and energy, while whole cottonseed, widely available in Texas, provides both fat and bypass protein. Distillers grains and soybean meal cover remaining protein needs. A mineral and vitamin premix and sodium bicarbonate as a rumen buffer round out the ration.

Target: 16-17% CP, 28-32% NDF, 25-30% starch, NEL 1.62-1.70 Mcal/kg DM, DMI 24-28 kg/cow/day.

Sources: National Research Council (NRC), *Nutrient Requirements of Dairy Cattle*, 7th ed. (2001); Texas A&M; AgriLife Extension, *Dairy Cattle Nutrition*; Shaver, R.D., *University of Wisconsin Extension, Corn Silage in Dairy Cattle Diets* (2020); Drouillard, J.S., *Kansas State University, Distillers Grains in Livestock Diets* (2018).

3. Key Model Assumptions

Variable	Value	Source / Justification
Herd size	100 lactating cows	Hypothetical mid-size Texas dairy
Milk yield	21.5 kg/cow/day	Conservative; US avg 28 kg/day (USDA NASS 2023)
Water use	150 L/cow/day	High end for TX heat; USDA range 100-200 L
Feed DMI	25 kg DM/cow/day	NRC 2001 range: 24-28 kg/day

Energy use	65 kWh/cow/month	USDA ERS 2023: 60-70 kWh/cow/month
Manure production	68 kg wet/cow/day	USDA NRCS; ~14% dry matter
Biogas yield	0.025 m ³ CH ₄ /kg manure	EPA AgSTAR: 20-30 L CH ₄ /kg VS
Solar output	850 kWh/month	NREL PVWatts; 100 kW; Brazos County TX
Precipitation	75 mm/month	NOAA 30-yr normals; College Station TX
Wastewater recycling	75%	Anaerobic lagoon standard
Simulation period	100 months	Sufficient to observe stock-flow equilibria

All values are literature-derived defaults. Calibrate with actual farm data before use. | April 2026